

Fiber

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions.

1. a) Explain the difference between the general communication system and optical fiber communication system using block diagram. 3
- b) What is optical fiber? Show its typical configuration with necessary diagram. 2
- c) Define the terms with suitable diagram: Critical angle, Phase velocity, Group velocity. 3
- d) Graphically show different ray propagation techniques in optical fiber. 2
- e) Develop two equations for numerical aperture that measure the light collecting ability of a fiber and show how the refractive index may be related to the numerical aperture. 4

2. a) Why the present form of optical communication is limited within the regions of 870nm, 1330nm and 1550nm? Explain with appropriate technical reasons. 4
- b) A step index fiber with a large core diameter compared with the wavelength of the transmitted light has an acceptance angle in air of 22° and a relative refractive index difference of 3%. Estimate the numerical aperture and the critical angle at the core-cladding interface for the fiber. 5 $n_1 \sqrt{2\Delta}$
- c) What are the important factors to be considered for making a suitable joint between two fibers? 2
- d) What is group index? Draw basic structure of an optical receiver. 3

3. a) What are different dispersion mechanisms in fiber optic waveguide? Derive the mathematical expression for the material dispersion parameter. 4
- b) Suppose the digital bit pattern 1011 of the broadening of light pulses as they are transmitted along a fiber as fiber input. Compare the fiber output at a distance for L_1 and L_2 ($L_1 < L_2$) 3
- c) What are the sources of losses in silica fiber optic? Explain in details the intrinsic absorption loss mechanisms with necessary illustrations. Also discuss fiber bending loss with proper diagram. 5
- d) Show different misalignment configurations that may occurs during connections between the fibers. 2

4. a) Describe the preparation method of optical fiber using liquid phase techniques. 4
- b) Write down the names of optical splices and connectors. Explain the applications of expanded beam connector. 4

Electrom
Rayleigh

- ✓ 5. Two step index fibers exhibit the following parameters:
- (i) A multimode fiber with a core refractive index of 1.5, a relative refractive index difference of 3% and an operating wavelength of $0.82\mu\text{m}$;
 - (ii) An $8\mu\text{m}$ core diameter single-mode fiber with a core refractive index the same as (i), a relative refractive index difference of 0.3% and an operating wavelength of $1.55\mu\text{m}$.

Estimate the critical radius of curvature at which large bending losses occur in both cases.

5. a) What are the different buffering techniques? Explain the Fresnel reflection mechanisms in an optical fiber joint. 3
- b) What is hetero-junction? Explain how a concentrated beam of light is obtained by double hetero-structure of optical devices. 3
- c) What are the advantages of LED? Show the basic structure of Laser and then explain the principles of light amplification in Laser cavity. 3
- d) Explain why does the responsivity of a photo detector become shorter wavelengths and the beyond cutoff wavelength. 2
- e) What do you mean by population inversion and pumping? 3
6. a) What is photo detector? Explain the photo detection principles in intrinsic material in brief. 3
- b) Show that the responsivity is directly proportional to the quantum efficiency at a particular wavelength. 3
- c) A p-n photodiode has a quantum efficiency of 50% at a wavelength of $0.9\mu\text{m}$. Calculate: 3
- (a) its responsivity at $0.9\mu\text{m}$;
 - (b) the received optical power if the mean photocurrent is 10^{-6}A ;
 - (c) the corresponding number of received photons at this wavelength.
- d) "The frequency response for an optical fiber system is more in optical than electrical bandwidth"- Justify the statement with relevant mathematical formula. 3
- e) Show the performance comparison of ELED and SLED. Draw the V/I characteristics of PD including different regions of operation. 2

Patuakhali Science and Technology University

Final Examination of B.Sc. Engg (CSE) Level-3, Semester-2, July-December-2015

Course Code: CCE-323 Course Title: Optical Fiber Communication

Session :2012-2013 Credit Hour: 3.00 Full Marks: 70 Duration: 3.00 Hours

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]

Answer any 7 of the following questions.

1. a) Show the basic structure of an optical fiber. 2
b) Illustrate different types of modulation technique. 3
c) Compare between power efficiency and peak to average power ratio. 3
d) Make differentiate between spectral efficiency and reliability. 2
2. a) Define photo detector. What is the difference between p-i-n photo detector and avalanched photodiode? 4
b) Distinguish among shot noise, thermal noise and amplified spontaneous emission noise. 3
c) Compare LED and semiconductor laser diode. 3
3. a) Define optical fiber sensor. What is the significance of optical fiber sensor? 3
b) Show different types of fiber connector. 2
c) Briefly describe different types of optical fiber sensor. 3
d) Write short notes on fiber coupler, circulator and mode scrambler. 2
4. a) Show the drawing and preparation of optical Fiber. 3
b) Illustrate different stages of optical fiber cabling with difference of dual core and six core fiber cables. 3
c) Draw the outdoor and undersea cable design of optical fiber. 2
d) Define fiber joint and joint loss. 2
5. a) Why optical amplification is needed? Write the general application of optical amplification. 3
b) Write down the issues of amplifier design. Show the categories of EDFA. 4
c) What is erbium doped fiber amplifier? Draw the basic EDFA amplifier design. 3
6. a) Differentiate between step index and graded index fiber. 3
b) Show the comparison between mathematical expression of material dispersion and waveguide dispersion. 2
c) Define the sources of noise of an optical receiver. 2
d) Distinguish between optical direct detection receiver and coherent detection receiver. 3
7. a) Briefly define the Raman amplifier. 3
b) Describe semiconductor optical amplifier (SOA) with its advantage and disadvantage. 3
c) Narrate the semiconductor optical amplifier (SOA) operation How WDM optical network works? 4
8. a) Write down the advantages and disadvantages of optical transmission. 2
b) Show the fiber distributed data interface (FDDI) specification with ring structure. 3
c) What do you mean by optical cross connect? Explain technologies for building a switch fabric of an OXC 3
d) How optical Add-Drop Multiplexer work. 2

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E3

Patuakhali Science and Technology University

6th Semester (L-3, S-II), Final Exam. of B. Sc. Engg. (CSE), July -Dec. 2016

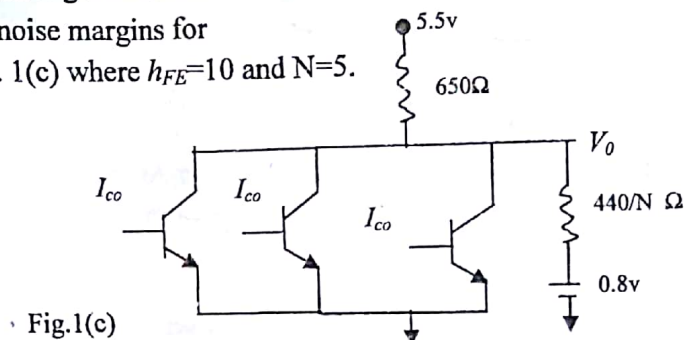
Course Code: EEE 321 Course Title: Digital Electronics and Pulse Technique

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

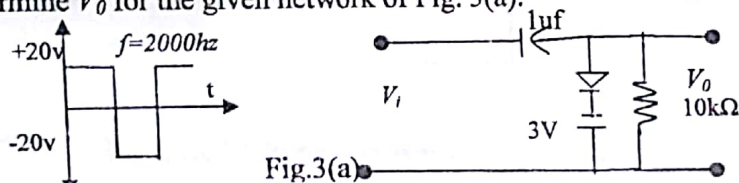
Answer any 5 of the following questions:

- ① (a). Draw a 3-input DTL NAND gate. Show the performance of DTL NAND gate based on your drawn circuit. [Considering standard values of diode and transistor parameters; $h_{FE}=40$] 6
- (b). Discuss about the saturated bipolar logic families. 4
- (c). Find out the logic 0 and logic 1 noise margins for the given circuit diagram of Fig. 1(c) where $h_{FE}=10$ and $N=5$. 4

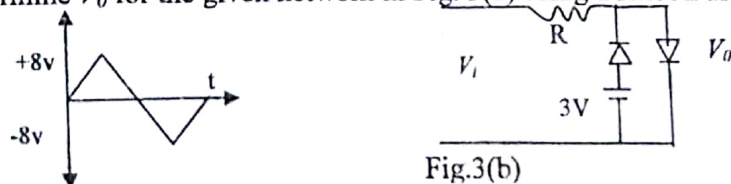


- ② (a). Why TTL is the most popular general purpose logic family? 3
- (b). If the digital input to the D/A converter is $\bar{b}_3b_2b_1b_0$, find the analog output voltage such that $V=0$ for a digital input of 0000. Design a D/A converter for the mentioned criteria. 3+4
- (c). How quantization error can be reduced in A/D converter? 4

- ③ (a). Determine V_O for the given network of Fig. 3(a). 6

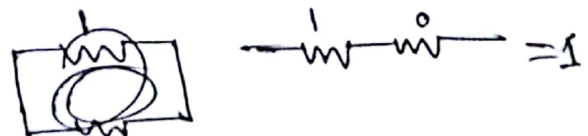


- (b). Determine V_O for the given network in Fig. 3(b) using a silicon diode. 5

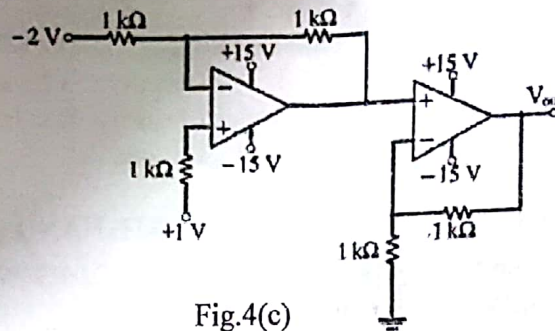


- (c). Distinguish series and parallel clippers. 3

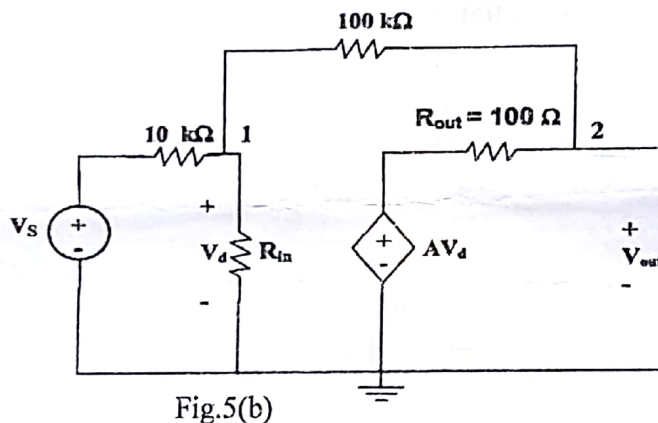
- ④ (a). What is an operational amplifier? "The inverting amplifier provides a constant voltage gain with 180° phase shift"- justify the statement with relevant circuit diagram. 5
- (b). Define slew rate and active device. Solve the differential equation using appropriate circuit diagram: $D^3y - 4D^2y + 6Dy - 8y = f(t)$ volts where $D = d/dt$ 5



- (c) Calculate the output voltage (V_{out}) for the circuit of fig.4(c)



5. (a). Write down the advantage of active filter. Design a 70 kHz with 50% duty cycle square wave generator using 555. 5
- (b). Sketch the equivalent circuit of an ideal Op-Amps. Suppose for a non-inverting amplifier, the equivalent circuit of an Op-Amps is shown in figure 5(b). Find: i) V_s/V_{out} for $R_{in}=100k\Omega$. 9
- ii) V_s/V_{out} for an ideal op-amp.



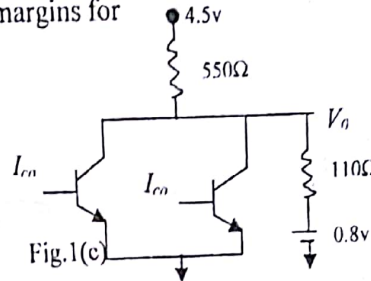
6. (a). What do you mean by cut off frequency? Show that in case of 1st order high pass filter, the transfer function is equal to the 70.7% of it's voltage gain. 5
- (b). "Monostable multivibrator is known as one-shot circuit" – Justify the statement with relevant circuit diagram. 4
- (c). Explain the functional block diagram of 555 timer. 5

WAN

A	B	C	OU
1	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

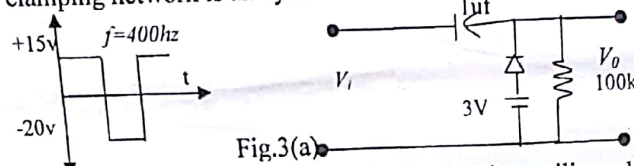
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1. (a). Explain the operation of a 3-input RTL OR gate with its circuit diagram. 6
 (b). Write down the short note on i. Figure of merit ii. Noise Margins. 4
 (c). Find out the logic 0 and logic 1 noise margins for the given circuit diagram of Fig. 1(c). 4

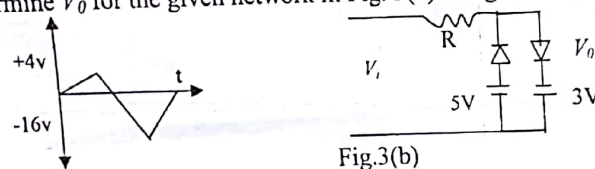


2. (a). Derive the equation of V_0 for weighted resistor D/A converter. 5
 (b). What are the problems of weighted resistor D/A converter? 3
 (c). Consider a 3-bit digital input in 1's complement format. Design a D/A converter for this. 6

3. (a). How clamping network is analyzed? Determine V_0 for the given network of Fig. 3(a). 9



- (b). Determine V_0 for the given network in Fig. 3(b) using a silicon diode with $V_T = 0.7V$; 5



4. (a). What is an operational amplifier? What are the fundamental differences between ideal & practical Op-Amp (Model 741c)? 4

- (b). Solve the differential equation using appropriate circuit diagram: 4

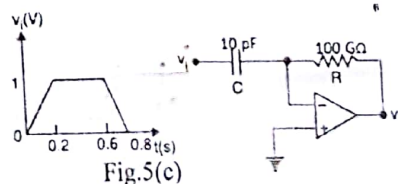
$$D^3y + 4D^2y - 6Dy - 8y = -5 \text{ volts where } D = \frac{d}{dt}$$

- (c). Write down the short notes on: i. Op-Amp is a subtractor ii. Frequency scaling iii. Schmitt trigger. 6

5. (a). Design a 50 kHz, 70% duty cycle square wave generator using 555 timer. 5

- (b). Show that in case of 1st order low pass filter, the transfer function is equal to the 70.7% of it's voltage gain. 5

- (c). For the differentiator circuit in the Fig.5(c), determine the output voltage and sketch the output waveform. 4



6. (a). Evaluate the time period for a free running multivibrator & show that the frequency is independent of output voltage of the square wave. 5

- (b). "Monostable multivibrator is known as one-shot circuit" – Justify the statement with relevant circuit diagram. 4

- (c). Explain the functional block diagram of 555 timer. 5

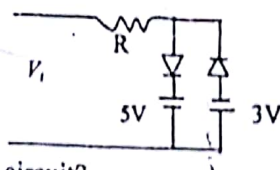
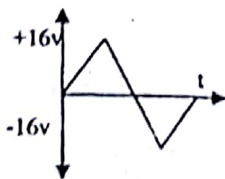
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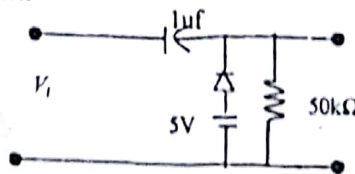
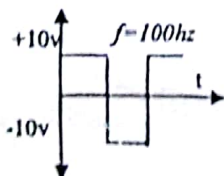
[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions:

1. (a). Explain the operation of a 3-input TTL NAND gate with its circuit diagram. 6
(b). Explain the operation of 2-input CMOS OR gate 4
(c). In the RTL 2-input NOR gate, $V_{cc} = 5V$, all resistances are 640Ω . Calculate the average power supply when it is driving 4 gates. Assume $V_{BE, Sat} = 0.8V$, $V_{CS, Sat} = 0.2V$, $h_{FE} = 20$. 4
2. (a). Discuss the operations of a monostable multivibrator with its circuit diagram where a trigger input is given at a NAND gate. 7
(b). Draw a circuit diagram for a comparator where reference voltage $V_{ref} = 5V$ and input voltage $V_{in} = 10 \sin 4\pi t$. Show the input and output waveforms for the given circuit. 7
3. (a). Draw the functional block diagram of 555 timer. Explain its operations. 8
(b). Discuss the function of a monostable multivibrator using 555 timer 6
4. (a). Design a D/A converter with $V(1) = -2V$, $V(0) = 0V$ and $R_F = 4R$ for 5 bit digital input from 00000 to 11111 7
(b). Adjust the offset voltage for the D/A converter such that $V_0 = 0$ for digital input of 01000. 7
5. (a). Design a 3-bit parallel comparator A/D converter. Explain the operation of this A/D converter. 8
(b). Explain how the quantization error can be reduced for this A/D converter. 6
6. (a). Determine V_0 for the given network using a silicon diode with $V_T = 0.7V$; 6



- (b). Why clamper is used in the electronic circuit?
Determine V_0 for the given network



[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions.

"An operating system is similar to a government."- Justify.

Do timesharing differs from multiprogramming? If so, How?

Differentiate between multiprogramming and multiprocessing with appropriate examples.

With a neat sketch illustrate the architecture of a Linux system.

"Context switch time is pure overhead"- Justify.

Contrast between preemptive and non-preemptive scheduling with appropriate examples.

Elucidate the queuing diagram representation of process scheduling.

How is a new process created using fork() system call? How many times does the following program print 'Operating System'?

```
#include<stdio.h>
```

```
#include<unistd.h>
```

```
int value=5;
```

```
int main()
```

```
{ int i;
```

```
for(i=0;i<3;i++)
```

```
fork();
```

```
printf("Operating System\n");
```

```
return 0;}
```

(a) What criteria affects the schedulers performance? Which ones should be maximized and minimized?

(b) Illustrate how a smaller time quantum increases context switches with an appropriate example.

(c) Explain the FCFS, preemptive and non-preemptive version of Shortest-Job-First and Round Robin (time slice = 2) scheduling algorithms with Gantt chart for the four processes given. Compare their average waiting time and turnaround time.

Process	Arrival Time	Burst Time	Process	Arrival Time	Burst Time
P ₁	0.0	7	P ₃	4.0	1
P ₂	2.0	4	P ₄	5.0	4

(a) What do you mean by deadlock? What four conditions must be fulfilled for a deadlock to occur?

(b) Suppose you have processes P₁, P₂, P₃ and resources R₁, R₂, R₃ and R₄. Draw the Resource allocation graph for the conditions such that P₁ has R₂ and requests for R₁, P₂ has R₁, R₂ and requests for R₃ and P₃ has R₃, R₂ and requests for R₂. The resources are single instance resources. Is the above situation a deadlock? Justify your answer.

(c) Consider a system with 12 magnetic tape drives and three processes P₁, P₂, P₃. Process P₁ requires 10 tape drives, process P₂ may need as many as 4 tape drives, and process P₃ may need up to 9 tape drives. Suppose that at a time t₁, process P₁ is holding 5 tape drives, process P₂ is holding 2 tape drives and process P₃ is holding 2 tape drives.

i. Is the system in safe state at time t₁? Why or why not?

ii. Suppose that at time t₂, process P₂ requests and is allocated one more tape drive. What will be the safety situation?

(a) "The page sizes are always power of 2" - Justify.

(b) A parent may terminate the execution of one of it's children. What are the reasons?

(c) Mention three requirements that a solution to the critical section problem must satisfy. Briefly describe the Peterson's solution to this problem.

(d) Given five memory partitions of 100 KB, 500 KB, 150 KB, 350 KB and 600 KB (in order), how would each of the first-fit, best-fit & worst-fit algorithms place processes of 110 KB, 419 KB, 210 KB and 423 KB (in order)? Which algorithm makes the most efficient use of memory and why?

(a) Distinguish between internal and external fragmentation.

(b) With a neat sketch illustrate how TLB is used with page table in paging scheme.

(c) Explain the RAID structure in disk management with various RAID levels of organization in detail.

(d) A small computer has four page frames. A process makes the following list of page references : 1,2,3,4,5,3,4,1,6,7,8,7,4,7,9,3. How many page faults would occur using Optimal, FIFO and LRU page replacement algorithms? Which algorithm results in minimum number of page faults and why?

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]
 Answer any 7 of the following questions.

1. (a) Define operating system. Justify the statement "An operating system is similar to a government." 03
- (b) Contrast between a process and a thread. Mention some benefits of using threads. 03
- (c) What is system program? Explain how OS handles a user application invoking the open() system call with relevant figure. 04
2. (a) What is context switch? "Context switch time is pure overhead"-justify the statement. 02
- (b) Explicate the queuing diagram representation of process scheduling. 04
- (c) How is a new process created using fork() system call? Using the program shown in Fig. 2(c), explain what the output will be at Line A. 04

```
#include<sys/types.h>      if (pid == 0)
#include<stdio.h>          { value+=15;
#include<unistd.h>          return 0; }
int value=5;               else if (pid > 0)
int main()                 { wait(NULL);
{                           printf ("Parent: value=%d", value); /*LINE A*/
    Pid_t=pid;              return 0; }}
    pid=fork();
```

Fig. 2 (c): What output will be at Line A?

3. (a) What criteria are used to measure the performance of a CPU scheduling algorithm? 03
- (b) Consider the following set of processes with the length of the CPU-burst time given in milliseconds: 07

Process	Arrival Time	Burst Time	Priority
P1	0.0	7-2	3
P2	2.0	4-3	2
P3	4.0	1 -	4
P4	5.0	4 -	1

- i. Draw two Gantt charts illustrating the execution of these processes using a preemptive priority (a smaller priority number implies a higher priority) & RR (time quantum=2) scheduling algorithms.
- ii. Compute what the average turnaround time will be if the CPU is left idle for the first 2 unit and then SJF scheduling is used.

4. (a) "A cycle in the graph is a necessary but not a sufficient condition for the existence of a deadlock"-justify the statement with an appropriate example. 02
- (b) What do you mean by deadlock? Depict a resource-allocation graph for the following situations: 04

2
1

$P = \{P_1, P_2, P_3\}$, $R = \{R_1, R_2, R_3, R_4\}$ and $E = \{P_1 \rightarrow R_1, P_2 \rightarrow R_3, R_1 \rightarrow P_2, R_2 \rightarrow P_2, R_2 \rightarrow P_1, R_3 \rightarrow P_3, P_3 \rightarrow R_2\}$

Resource Instances: R_1 (1), R_2 (2), R_3 (1), R_4 (3). And figure out the cycle if it does contain in the graph.

- (c) Consider a system with 12 magnetic tape drives and three processes P_1, P_2, P_3 . Process P_1 requires 10 tape drives, process P_2 may need as many as 4 tape drives, and process P_3 may need up to 9 tape drives. Suppose that at a time t_1 , process P_1 is holding 5 tape drives, process P_2 is holding 2 tape drives and process P_3 is holding 2 tape drives. 04

- Is the system in safe state at time t_1 ? Why or why not?
- Suppose that at time t_2 , process P_2 requests and is allocated one more tape drive. What will be the safety situation?

- (a) What four conditions must be fulfilled for a deadlock to occur? 02
- (b) Semaphore S is an integer variable which is manipulated through two operations, $\text{wait}()$ and $\text{signal}()$. 03

Definition of $\text{wait}()$:

```
wait(S){
    while S <= 0;
```

Definition of $\text{signal}()$:

```
signal(S){
    S++;
```

How can you overcome the semaphore definition problem by modifying the definition of $\text{wait}()$ and $\text{signal}()$ operation? block make up

- (c) Mention three requirements that a solution to the critical section problem must satisfy. Briefly describe the Peterson's solution to this problem. 05

- (a) "The page sizes are always power of 2" – justify the statement. 02

- (b) State the solution to external fragmentation problem. Explain the segmentation hardware along with an appropriate example. 04

- (c) A small computer has four page frames. A process makes the following list of page references : 1,2,3,4,1,2,5,1,2,3,4,5. 04

How many page faults would occur using Optimal, FIFO and LRU page replacement algorithms? Which algorithm results in minimum number of page faults and why?

- (a) What is the meaning of the term busy waiting? 02

- (b) What is the advantage of using TLB in paging? Illustrate how TLB is used with page table in paging scheme. 04

- (c) Given five memory partitions of 100 KB, 500 KB, 150 KB, 350 KB & 600 KB (in-order), how would each of the first-fit, best-fit & worst-fit algorithms place processes of 417 KB, 110 KB, 212 KB & 425 KB (in-order)? Which algorithm makes the most efficient use of memory and why? 04

8. (a) What do you mean by dual mode operation of an operating system? 02

- (b) Explicate the Consumer -Producer problem. Where is the critical section of this problem? 04

- (c) Consider the exponential averaging method is used to estimate the length of next CPU burst for a process. Assume $\pi_0 = 10$, $\alpha = 0.2$ and the sequence of really observed CPU burst length so far is: 12, 18, 4, 10, 16. What is the sequence of estimated CPU burst length for next CPU burst? 04

9. (a) Define indefinite postponement. 02

- (b) "Context switching time is high for swapping" – Justify the statement with an appropriate example. 04

- (c) Consider a system with 80% hit ratio, 200 nano-seconds time to access memory & 20 nanoseconds time to access TLB. Calculate the Effective Access Time (EAT). 04

05

Patuakhali Science and Technology University
B.Sc. Engg. (CSE) Level-3, Semester-II, Final Examination Jul-Dec/14, Session: 2011-12
Course code: CIT-321 **Course Title: Operating System**
Credit hour: 3.00 **Full marks: 70** **Duration: 3 hours**
 [Figures in the right margin indicate full marks.]

Answer any 7 of the following questions. Split answering is not recommended.

1. ☒ a. How does an OS work as a resource allocator? 3
☒ b. Why is Direct Memory Access (DMA) used? 3
☐ c. What are the advantages of multiprocessors systems? 4
 2. ☒ a. What is a system call? Why should we use APIs rather than system calls? 4
☐ b. Suppose you want to copy a file's content into a newly created file. What will be the systems calls for this purpose? 6
 3. ☒ a. Discuss OS services. 5
☐ b. Describe different methods to pass parameters to the OS. 5
 4. ☒ a. Define process. What are the different parts of a process? 4
☐ b. What are different process states? Draw the possible state diagram for a process. 6
 5. ☒ a. Discuss different types of process scheduler. 7
☐ b. Why should the CPU-scheduler be very fast? 3
 6. ☒ a. Suppose you have processes P_0, P_1, P_2, P_3, P_4 and Resources R_0, R_1, R_2, R_3, R_4 . Draw the resource allocation graph for the conditions such that P_0 has R_0 and requests for R_1 and R_3 , P_1 has R_4 and requests for R_3 and R_2 , P_2 has R_1 and requests for R_4 , P_3 has R_3 and requests for R_0, P_1 and R_4 , P_4 has R_2 and requests for R_4 . The resources are single instance resources. 6
☐ b. Is the above situation a deadlock? Justify your answer. 4
 7. ☒ a. What are the issues that should be taken care of during preemptive scheduling? 4
☐ b. What are the criteria for CPU scheduling? Which ones should be maximized and minimized? 6
 8. ☒ For the following processes with the given burst time and arrival time, how can the processes be scheduled using SJF preemptive, SJF nonpreemptive, RR scheduling ($Q=3ms$)? Draw the Gantt chart. Calculate average waiting time for the processes. 10
- | Process | Arrival Time | Burst Time (ms) |
|---------|--------------|-----------------|
| P_1 | 0 | 5 |
| P_2 | 2 | 3 |
| P_3 | 3 | 2 |
| P_4 | 5 | 9 |
| P_5 | 9 | 1 |
9. Suppose the memory has 6 free partitions of 5KB each. 5 Processes arrives as it is given in Question 8. The processes have 7KB, 4KB, 10KB, 8KB and 1KB memory requirements respectively. If it is a nonpreemptive case, how will memory be allocated to the processes using First-fit, Best-fit and Worst-fit algorithms? Calculate internal and external fragmentations for the algorithms. 10

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions:

1. (a). Characterize continuous systems and discrete systems in practice. 5
 (b). Deduce T_i , Area under $Q(t)$, Area under $B(t)$ for single server queuing system in computer representation with the parameters of $A_1=0.3$, $A_2=1.0$, $A_3=0.5$, $A_4=1.5$, $A_5=0.8$ and $S_1=1.3$, $S_2=1.0$, $S_3=0.5$, $S_4=1.0$. Draw the chart of $Q(t)$ and $B(t)$ for given parameters. 9
2. (a). Show the ways to study a system. 3
 (b). Generate a sequence of 50 random numbers using arithmetic congruential generator between 10 and 70 where $r_1=9$, $r_2=11$, $m=13$. Test their uniformity employing the Chi-Squared test where the accepted value of Chi-square at 95% confidence level is 11.07. 6
3. (a). If x is a random variable with the following distributions $f(x) = 1.4xe^{-0.4x}$ show graphically the p.d.f. 5
 (b). Three fair dices are rolled once. The sum of the three numbers on the dice represents the outcomes for a random variable x . Determine the probability distribution when x is even one. 7
 (c). What are conditions of discrete probability functions? 2
4. (a). What is Monte Carlo Method? What is its application? Determine the numerical integration by Monte Carlo method with flowchart. 5
 (b). Discuss the development process for system simulation model. 6
 (c). What are the techniques to be used to debug any computer program of a simulation modelling? 3
5. (a). Suppose we have $n = 40$ patients who will be receiving an experimental therapy which is believed to be better than current treatments which historically have had a 5-year survival rate of 20%, i.e. the probability of 5-year survival is $p = .20$. Using the new treatment we find that 16 out of the 40 patients survive at least 5 years past diagnosis.
 Does this result suggest that the new therapy has a better 5-year survival rate than the current, i.e. is the probability that a patient survives at least 5 years greater than .20 or a 20% chance when treated using the new therapy? 5
 (b). Explain the following with an example: 4
 Attribute, Entity, Activity, and State of the system.
 (c). Give the difference between simulation and analytical methods with example. What are the purposes for using simulation? 4
 (d). Explain $E_k / D / 10 / 2 / GD$ 2
6. (a). State and explain Poisson Distribution. Generate five random observations from a Poisson distribution, if the mean is 4.0. 4
 (b). Discuss three step approach for developing valid and credible simulation models. 3
 (c). Write the desirable features of a simulation software. 4
 (d). Which of the following is most likely to be well modelled by a Poisson distribution? Why? 3
 - i. Number of trains arriving at Kamlapur every hour
 - ii. Number of lottery winners each year that live in Barisal
 - iii. Number of days between solar eclipses
 - iv. Number of days until a component fails

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions:

1. (a). Discuss whether the simulation should be (i) deterministic or stochastic (ii) continuous or discrete. 5
 (b). Show the flow chart for the next-event time advance approach. 5
 (c). For a SSQ, arrival occurs at times 0.1, 0.5, 1.1, 1.3, 2.1, 3.4, 3.7 and departure occurs at times 0.9, 1.4, 2.5, 3.3 and 4.4. The simulation ends at time $T(6)=4.4$. Show the graph of $Q(t)$ and compute the values of T_i and $\bar{q}(n)$. 4
2. In the SSQ, $A_1=0.2$, $A_2=1.0$, $A_3=0.2$, $A_4=1.2$, $A_5=0.5$ and $S_1=0.5$, $S_2=1.7$, $S_3=1.2$, $S_4=1.0$; Show the computer representation with discussion for the first five events. 14
3. (a). Generate a sequence of 10 random numbers between 0 and 1 using the multiplicative congruential method where $r_0=89$, $a=5$; $m=77$. 6
 (b). The following sequences of random numbers are generated as 0.24, 0.55, 0.33, 0.22, 0.86, 0.55, 0.39, 0.59, 0.77 and 0.58. Use the Kolmogorov-smirnov test for the uniformity test of given random number where the critical value is 0.41. 8
4. (a). What are the limitations of mid-square random number generator? Generate 50 random numbers using mid-square algorithms. 7
 (b). Test the acceptance of random number through Chi-squared method where the accepted value of Chi-square is 9.448. 7
5. (a). The numbers of customers are counted for a service station as 3, 2, 6, 10, 8, 1, 5, 3, 2, 5, 7, 1, 3, 4, 3, 8, 9, 2, 3, 6, 10. Determine the mean, mode, median, standard deviation and coefficient of variation. 8
 (b). The sum of the two numbers on the dice represent the outcome for a random variable x . Determine the probability distribution of x . What is the probability when x is even? 6
6. (a). Write down the density function, mean, variance and mode of Uniform distribution, Exponential distribution, Gamma distribution and Normal distribution with the graphical representation. 8
 (b). Show the mass function of Burnoulli distribution and Binomial distribution with their graphical representations. 6

2016

16

Patuakhali Science and Technology University
6th Semester (L-3, S-II), Final Exam of B. Sc. Engg. (CSE), July -Dec. 2016
Course Code: CCE 321 Course Title: Computer Peripheral and Interfacing Session: 2013-2014
Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions:

Q. 1

1. (a). Differentiate between MDA and CGA. 2
(b). Write down the advantages and disadvantages of LCD over CRT display. 4
(c). When a key on a keyboard is pressed, a switch is activated to generate signals for communicating the same to the PC. Give a brief description of four of them. 8
2. (a). What is key debouncing? Why is it needed? 2
(b). Define make code, break code, typematic, persistence, aspect ratio and resolution. 3
(c). Modern keyboards support bidirectional data transfer. How? 5
(d). Explain the phase sequences for the execution of "Test Unit Ready" SCSI command. 4
3. (a). Why is serial ATA used in present technologies instead of parallel ATA? 3
(b). Describe IDE command execution protocols. 10
(c). What do you mean by "CHRN" value? 1
4. (a). Describe Data Read and Data Write operations through FDC. 5
(b). Encode data 101000100110 using FM, MFM and RLL encoding formats. Show the waveforms. 9
5. (a). Illustrate an 8-bit microprocessor interfaced to a 4K RAM system using a full/partial decoded addressing technique. Also show the address map corresponding to the above mentioned organization. 5
(b). What do you mean by ISA bus and PCI bus? 2
(c). Write down the steps involved in starting a printing sequence using a printer controller chip. 4
(d). Explain the operation of a delta-delta shadow-mask CRT. 3
6. (a). What are the basic differences between ASCII and EBCDIC? 2
(b). Describe the process of a device initiated I/O transfer. 3
(c). When is an overflow flag set or reset? Give examples? 3
(d). Illustrate the character and signal generation procedure of a CRT display with a short description. 4
(e). The MMU can perform address translation in one of the two ways. Define them. 2

Memory

Patuakhali Science and Technology University

B.Sc.Engg.(CSE) L-3, S-2, Final Examination -2015 (July-Dec). Session: 2012-13

Course Code: CCE-321

Course Title: Computer Peripheral and Interfacing

Credit Hour: 3.0

Full Marks: 70

Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

[Use figures where necessary]

Answer any 5 of the following questions:

P. 10

1. (a). What is buffer? Why we use buffers? 3
- (b). Define interfaces. What are the characteristics of computer peripherals? 4
- (c). Differentiate between buffers and spoolers. 2
- (d). What do you mean by solid state storage device? Write down the features and advantages of Solid state storage devices. 5
2. (a). Define raster scan display and random scan display. What are the differences between raster scan system and random scan system? 5
- (b). What are the advantages of LCD panel over conventional CRT monitor? 4
- (c). Differentiate between ATA RAID and SCSI RAID. 2
- (d). Write down the selection criteria of a monitor. 3
3. (a). Define USB. What are the functions of USB ports? Write down the limitations of USB ports. 5
- (b). Differentiate between IEEE1394 and USB. 3
- (c). Sketch the three tier topology of USB. Write down the benefits of three tier topology of USB. 4
- (d). What are the benefits of using USB? 2
4. (a). What is PCI Express? Write down the features and applications of PCI Express bus. 7
- (b). Define resource conflict. How can user resolve resource conflict? 4
- (c). Differentiate between serial ATA and parallel ATA. 3
5. (a). Differentiate between ISA, VESA and EISA buses. 5
- (b). What do you mean by ATA? Write down the features of the ATA-1 specification. 4
- (c). Write short note on: AGP, Plug and Play, SCSI. 3
- (d). Differentiate between IRQ and DMA channels. 2
6. (a). Define RAID. Describe the different levels of RAID. 7
- (b). Differentiate between IDE/ATA and SCSI. 4
- (c). Write down the advantages of SCSI over ATA. 3

Patuakhali Science and Technology University

B.Sc. Engg. (CSE) L-3, S-2, Final Examination -2014 (July-Dec), Session: 2011-12
Course Code: CCE-321 Course Title: Computer Peripheral and Interfacing

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

[Use figures where necessary]

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4. (a). What is PCI Express? Write down the features and applications of PCI Express bus. 8
(b). Define resource conflict. How can user resolve resource conflict? 4
(c). What do you mean by voltage conversion and protocol conversion? 2
5. (a). Compare between VESA and PCI buses. 5
(b). What do you mean by ATA? Write down the features of the ATA-1 specification. 6
(c). Write short note on: AGP, Plug and Play, SCSI. 3
6. (a). Define RAID. Describe the different levels of RAID. 7
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